The meaning of the homotopy of higher categories

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What is an $(\infty,1)$-category

(∞,n)-category

What is a homotopy theory?

- Quillen Model category

$(M, \underset{\text{cofib}}{C}, \underset{\text{fib}}{F}, W)$ — weak equiv

- Quillen A-functor! — degree preserve all the structure

- weak, less structure

homotopical category

(C, W) — weak eq satisfy 2-out-of-6

$f \xrightarrow{g} h$

gf, hg ca $\Rightarrow$ all in W

Essentially category, half functor, can be extracted from weak equivalences

- Relative case — Mapping spaces between objs

(C, W) — subcat of "weak equiv"

all identities

$\mathrm{Rel}\,\mathrm{Cat} \xrightarrow{N^H} \mathrm{Cat}_\infty$

Hammock localization

L^H of nerve localization

grids of zig zags.

Model Cat $\longrightarrow$ Cat_∞

L^H

gives equiv equivalence simplicial cats.

Charles Rezk's Complete Segal Space

Htpy thy $\rightsquigarrow$ underlying space of objects X_0 (classifying space)

Space of self homotopies.

id — "fiber bundle" — fibers are htpy in htpy thy.

same space $Y \to X_0$ fiber m.

$\rightsquigarrow$ underlying space of maps. X_1

underlying of ...

$X_0 \leftleftarrows X_1 \rightleftarrows X_2 \cdots$

simplicial space

w/ axioms

Cat$_\infty$ $\xleftarrow{L^H}$ RelCat

(SS)

cofibrant cat — set

C — NC_0 = objs

NC_1 = maps

$NC_2 = \mathrm{Fun}([2], C)$

$\cdots \to ! \to !$

characterize

$NC_0 \to NC_1 \to NC_2$

$NC_2 \to NC_1$

$\downarrow \qquad \downarrow$

$NC_1 \to NC_0$

pullback square (in set)

Fun Facts: there are really 9 model structures on Set.

up to Quillen eq

3 different

1-1 have 1 structure in Set

An (∞,n)-cat.

"higher cat w/ morphisms at all levels"

+ invertible above n

Model structure of these capture the htpy thy

Bar — can say cats stuff $(\infty,n\text{-}1)$

$(\infty,1)$-cat = ?

mapping $(\infty,n-1)$-cat

invertible.

Htpy hypothesis

∞-groupoids = spaces

(htpy thy + ~)

n-groupoids = n-types

$K_{\mathbb{Z}}X \leftarrow X$

Segal category. set of objs C_0

$C_1 = \coprod_{x,y} C(x,y)$ — space of maps.

$C_2 = \coprod_{x,y,z} C(x,y,z)$

$\therefore$ get simplicial space.

Segal cond

$C_n \xrightarrow{\simeq} C_1 \times_{C_0} C_1 \times_{C_0} \cdots \times_{C_0} C_1$

htpy equiv.

weak composition

$C(x,y,z) \xrightarrow{\simeq} (x,y) \times (y,z)$

$C(x,z)$ — choose composition.

basically invertible space of compositions

2 model structures for Seg

$$\begin{tikzcd}
\mathrm{Seg} \arrow[r, "\mathrm{pos}"] & \mathrm{Cat}_\infty \arrow[d, shift right] & \mathrm{RelCat} \arrow[l, shift right, "L^H"'] \arrow[d, shift left] \\
\mathrm{Seg}(\infty) \arrow[u] \arrow[r] & \mathrm{CSS} \arrow[u, shift right] \arrow[l, shift right] &
\end{tikzcd}$$

Bousfield

Htpy theories — right Quillen equiv.

of all $(\infty,1)$-cats

similar thy for (∞,n)

$\boxtimes$ = cartesian

$\boxtimes$ = simplicial

m = model cat of $(\infty,n-1)$-cat

given a rel cat

models of (∞,n)-cats

$$\begin{tikzcd}
\mathrm{Cat}_\infty^m \arrow[d, shift right] & \mathrm{RelCat} \arrow[d, shift left] \\
\mathrm{Seg}_m \arrow[u, shift right] \arrow[d, shift right] & \mathrm{fSS}_m \arrow[u, shift left] \arrow[d, shift left] \\
\mathrm{Seg}_m^{(\infty)} \arrow[u, shift right] & \mathrm{CSS}_m \arrow[u, shift left]
\end{tikzcd}$$

$\|M\|$ — simplicial cats

$\|\mathrm{Top}\|$

nSeg Seg_m

Thm) If Axioms characterize the htpy thy (Quillen category)

of (∞,n)-cat up-to equiv

Thm

Space of quasicats satisfying these axioms is $(B\mathbb{Z}/2)^{n-1}$ — htpy-equiv homo. case

$(B\mathbb{Z}/2)$ — classifying Cat.

Thm (Nerdy) All these models of (∞,n)-cat satisfy axioms

$\mathrm{Aut}(L^H(SS)) \simeq \mathbb{Z}/2$

only $\mathbb{Z}/2$ automorphism (p)

Axioms

C presentable, have some htpy limit.

$\mathcal{T}_n \to C$

$C_{\mathcal{T}_n \in C}$ — $\mathrm{Hom}(\mathcal{T}_n, C) = C_{\mathcal{T}_n}$

(1) Every obj of C gen by cells under limits, colimits

$(\mathrm{colim}\ \mathcal{T}_n) = X$

$\mathcal{T}_n \to X$

(2) $C/C_{\mathcal{T}_n}$ have right adjoint (preserves limits)

(3) finite lim of objects colimits

(4) Universal property. w.r.t. 1-3

D same 1-3

$C \underset{R}{\overset{L}{\rightleftarrows}} D$

(n-1)-cat

can build the universal guy k

Def $\mathcal{T}_n$ strict n-cat.

smallest subcat of strict n-cat closed under.

$(-) \times_{C_0} (-)$ pullbacks

$\mathcal{T}_n \hookrightarrow C$

$\begin{pmatrix} \mathrm{colim}\ \mathcal{T}_n \\ \mathcal{T}_n \to X \end{pmatrix} = X$

Facts - cells do gen every thy under limit.

- If M, N model cats satisfy axioms

$L: M \rightleftarrows N: R$

Quillen adjunction.

If L preserves cells

$\Rightarrow (L, R)$ Quillen equiv

[Rezk]

Htpy Hypothesis satisfy by CinSp

k-groupoids in CinSp $\Leftrightarrow$ k-types

✓ small.

[Lurie] Cob Hypp in CSS_n

$\Downarrow$

✓ invert models.

can be used to characterize — a lot of stuff.

(∞,∞)-cat of (∞,n) cats ...?

(∞,n) cat of (∞,n) cats — determines

(some "internal Hom")